



Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Kinematics 7 Supplement: the four forms of

$$v^2 - u^2 = 2as \text{ — ANSWERS}$$

1. Order of operations — BIDMAS

Always remember BIDMAS!

	Stands for	What to do	Example
B	Brackets	calculate inside brackets first	$(3 + 2) \times 4 = 5 \times 4 = 20$
I	Indices (powers)	then do exponents	$3 + 4^2 = 3 + 16 = 19$
DM	Division, Multiplication	then $\div$ or $\times$, left to right	$12 \div 3 \times 2 = 4 \times 2 = 8$
AS	Addition, Subtraction	then $+$ or $-$, left to right	$10 - 3 + 2 = 7 + 2 = 9$

More Tips:

- **FRACTIONS** - consider these like Brackets. Do the top calculations, then the bottom calculations, then divide the whole thing.
- **SQUARE ROOT** - A square root sign works like brackets. Work out everything under the $\sqrt{\quad}$, then calculate the $\sqrt{\quad}$ after.

2. SUVAT — what each letter means

Symbol	Name	Unit	Unit in words
s	displacement (distance)	m	metres
u	initial velocity	m/s	metres per second
v	final velocity	m/s	metres per second
a	acceleration	m/s ²	metres per second squared
t	time	s	seconds

A. Find the distance, s — form A: $s = \frac{v^2 - u^2}{2a}$

Worked example A — find the distance

A bus speeds up from 4.0 m/s to 12 m/s with an acceleration of 2.0 m/s². Calculate the distance it travels while speeding up.

What the question tells you

Initial velocity = u = 4.0 m/s

Final velocity = v = 12 m/s

Acceleration = a = 2.0 m/s²

Distance = s = ? *the one the question asks for*

Time = t = *not in the question: circle t in the Givens and cross it out*

1. Givens

$s = ?$

$u = 4.0$ m/s

$v = 12$ m/s

$a = 2.0$ m/s²

~~t~~ =

2. Equation $v^2 - u^2 = 2 a s$

3. Substitute $12^2 - 4.0^2 = 2 \times 2.0 \times s$

4. Rearrange $s = \frac{12^2 - 4.0^2}{2 \times 2.0}$ *form A*

5. Calculate $v^2 = 12^2 = 144$

$u^2 = 4.0^2 = 16$

$v^2 - u^2 = 144 - 16 = 128$

$2 \times a = 2 \times 2.0 = 4.0$

$s = 128 \div 4.0 = \mathbf{32\ m}$

Check: $v^2 - u^2 = 128$ and $2 a s = 2 \times 2.0 \times 32 = 128$. Same number.

Now try. Fill in the Givens, circle and cross out t , then work down the ladder, one line at a time.

A1. A cyclist speeds up from rest to 6.0 m/s with an acceleration of 1.5 m/s². Calculate the distance.

Givens

$s = ?$

$u = 0$

$v = 6.0$ m/s

$a = 1.5$ m/s²

~~t~~ =

Form: $s = \frac{v^2 - u^2}{2a}$

$v^2 = 6.0^2 = 36$

$u^2 = 0^2 = 0$

$v^2 - u^2 = 36 - 0 = 36$

$2 \times a = 2 \times 1.5 = 3.0$

$s = 36 \div 3.0 = \mathbf{12\ m}$

Check: $v^2 - u^2 = 36$ $2 a s = 36$

A2. A car speeds up from 10 m/s to 20 m/s with an acceleration of 3.0 m/s². Calculate the distance.

Givens

$$\begin{aligned} s &= ? \\ u &= 10 \text{ m/s} \\ v &= 20 \text{ m/s} \\ a &= 3.0 \text{ m/s}^2 \\ \cancel{\times} &= \end{aligned}$$

$$\begin{aligned} \text{Form: } s &= \frac{v^2 - u^2}{2a} \\ v^2 &= 20^2 = 400 \\ u^2 &= 10^2 = 100 \\ v^2 - u^2 &= 400 - 100 = 300 \\ 2 \times a &= 2 \times 3.0 = 6.0 \\ s &= 300 \div 6.0 = 50 \text{ m} \end{aligned}$$

Check: $v^2 - u^2 = 300$ $2as = 300$

B. Find the acceleration, a — form B: $a = \frac{v^2 - u^2}{2s}$

Worked example B — find the acceleration

A sprinter speeds up from 3.0 m/s to 9.0 m/s over a distance of 24 m. Calculate the sprinter's acceleration.

1. Givens

$$\begin{aligned} s &= 24 \text{ m} \\ u &= 3.0 \text{ m/s} \\ v &= 9.0 \text{ m/s} \\ a &= ? \\ \cancel{\times} &= \end{aligned}$$

2. Equation $v^2 - u^2 = 2as$

3. Substitute $9.0^2 - 3.0^2 = 2 \times a \times 24$

4. Rearrange $a = \frac{9.0^2 - 3.0^2}{2 \times 24}$ form B

5. Calculate $v^2 = 9.0^2 = 81$
 $u^2 = 3.0^2 = 9.0$
 $v^2 - u^2 = 81 - 9.0 = 72$

$$2 \times s = 2 \times 24 = 48$$

$$a = 72 \div 48 = 1.5 \text{ m/s}^2$$

Check: $v^2 - u^2 = 72$ and $2as = 2 \times 1.5 \times 24 = 72$. Same number.

Now try. In form B the bottom line is $2 \times s$.

B1. A scooter starts from rest and reaches 12 m/s over 36 m. Calculate its acceleration.

Givens

$$\begin{aligned} s &= 36 \text{ m} \\ u &= 0 \\ v &= 12 \text{ m/s} \\ a &= ? \\ \cancel{\times} &= \end{aligned}$$

$$\begin{aligned} \text{Form: } a &= \frac{v^2 - u^2}{2s} \\ v^2 &= 12^2 = 144 \\ u^2 &= 0^2 = 0 \\ v^2 - u^2 &= 144 - 0 = 144 \\ 2 \times s &= 2 \times 36 = 72 \\ a &= 144 \div 72 = 2.0 \text{ m/s}^2 \end{aligned}$$

Check: $v^2 - u^2 = 144$ $2as = 144$

B2. A bike slows from 8.0 m/s to a stop over 16 m. Calculate the acceleration. Slowing down gives a negative acceleration, so your answer has a minus sign.

Givens

$$\begin{aligned}s &= 16 \text{ m} \\ u &= 8.0 \text{ m/s} \\ v &= 0 \\ a &= ? \\ \cancel{\times} &= \end{aligned}$$

$$\text{Form: } a = \frac{v^2 - u^2}{2s}$$

$$v^2 = 0^2 = 0$$

$$u^2 = 8.0^2 = 64$$

$$v^2 - u^2 = 0 - 64 = -64$$

$$2 \times s = 2 \times 16 = 32$$

$$a = -64 \div 32 = -2.0 \text{ m/s}^2$$

Check: $v^2 - u^2 = -64$ $2as = -64$

C. Find the final velocity, v — form C: $v = \sqrt{u^2 + 2as}$

Worked example C — find the final velocity

A scooter moving at 9.0 m/s accelerates at 2.0 m/s² over a distance of 10 m. Calculate its final velocity.

1. Givens

$$\begin{aligned}s &= 10 \text{ m} \\ u &= 9.0 \text{ m/s} \\ v &= ? \\ a &= 2.0 \text{ m/s}^2 \\ \cancel{\times} &= \end{aligned}$$

2. Equation $v^2 - u^2 = 2as$

3. Substitute $v^2 - 9.0^2 = 2 \times 2.0 \times 10$

4. Rearrange $v = \sqrt{9.0^2 + 2 \times 2.0 \times 10}$ form C

5. Calculate $u^2 = 9.0^2 = 81$

$$2as = 2 \times 2.0 \times 10 = 40$$

$$v^2 = 81 + 40 = 121$$

$$v = \sqrt{121} = 11 \text{ m/s}$$

Check: $v^2 - u^2 = 121 - 81 = 40$ and $2as = 40$. Same number.

Now try. Add first, and press $\sqrt{}$ last.

C1. A go-kart starts from rest and accelerates at 2.0 m/s² over 36 m. Calculate its final velocity.

Givens

$$\begin{aligned}s &= 36 \text{ m} \\ u &= 0 \\ v &= ? \\ a &= 2.0 \text{ m/s}^2 \\ \cancel{\times} &= \end{aligned}$$

Form: $v = \sqrt{u^2 + 2as}$

$$u^2 = 0^2 = 0$$

$$2as = 2 \times 2.0 \times 36 = 144$$

$$v^2 = 0 + 144 = 144$$

$$v = \sqrt{144} = 12 \text{ m/s}$$

Check: $v^2 - u^2 = 144$ $2as = 144$

C2. A cyclist moving at 3.0 m/s accelerates at 3.0 m/s² over 12 m. Calculate the final velocity.

Givens

$$s = 12 \text{ m}$$

$$u = 3.0 \text{ m/s}$$

$$v = ?$$

$$a = 3.0 \text{ m/s}^2$$

$$\cancel{v} =$$

$$\text{Form: } v = \sqrt{u^2 + 2as}$$

$$u^2 = 3.0^2 = 9.0$$

$$2as = 2 \times 3.0 \times 12 = 72$$

$$v^2 = 9.0 + 72 = 81$$

$$v = \sqrt{81} = 9.0 \text{ m/s}$$

Check: $v^2 - u^2 = 72$ $2as = 72$

D. Find the initial velocity, u — form D: $u = \sqrt{v^2 - 2as}$

Worked example D — find the initial velocity

A motorbike accelerates at 3.0 m/s² over 20 m and reaches 13 m/s. Calculate how fast it was going at the start.

1. Givens

$$s = 20 \text{ m}$$

$$u = ?$$

$$v = 13 \text{ m/s}$$

$$a = 3.0 \text{ m/s}^2$$

$$\cancel{v} =$$

2. Equation $v^2 - u^2 = 2as$

3. Substitute $13^2 - u^2 = 2 \times 3.0 \times 20$

4. Rearrange $u = \sqrt{13^2 - 2 \times 3.0 \times 20}$ form D

5. Calculate $v^2 = 13^2 = 169$

$$2as = 2 \times 3.0 \times 20 = 120$$

$$u^2 = 169 - 120 = 49$$

$$u = \sqrt{49} = 7.0 \text{ m/s}$$

Check: $v^2 - u^2 = 169 - 49 = 120$ and $2as = 120$. Same number.

Now try. Subtract first, and press $\sqrt{}$ last.

D1. A car accelerates at 2.0 m/s² over 7.0 m and reaches 8.0 m/s. Calculate its initial velocity.

Givens

$$s = 7.0 \text{ m}$$

$$u = ?$$

$$v = 8.0 \text{ m/s}$$

$$a = 2.0 \text{ m/s}^2$$

$$\cancel{v} =$$

$$\text{Form: } u = \sqrt{v^2 - 2as}$$

$$v^2 = 8.0^2 = 64$$

$$2as = 2 \times 2.0 \times 7.0 = 28$$

$$u^2 = 64 - 28 = 36$$

$$u = \sqrt{36} = 6.0 \text{ m/s}$$

Check: $v^2 - u^2 = 28$ $2as = 28$

D2. A horse accelerates at 3.0 m/s^2 over 24 m and reaches 15 m/s . Calculate its initial velocity.

Givens

$$s = 24 \text{ m}$$

$$u = ?$$

$$v = 15 \text{ m/s}$$

$$a = 3.0 \text{ m/s}^2$$

$$\cancel{x} =$$

$$\text{Form: } u = \sqrt{v^2 - 2as}$$

$$v^2 = 15^2 = 225$$

$$2as = 2 \times 3.0 \times 24 = 144$$

$$u^2 = 225 - 144 = 81$$

$$u = \sqrt{81} = 9.0 \text{ m/s}$$

$$\text{Check: } v^2 - u^2 = 144 \quad 2as = 144$$

E. Calculator round — one of each form, in the same order; answers to 2 significant figures

Same steps: one calculator line at a time. Round only on the last line, and check with the number before you rounded.

E1. A runner speeds up from 2.5 m/s to 7.5 m/s at 0.60 m/s^2 . Calculate the distance.

Givens

$$s = ?$$

$$u = 2.5 \text{ m/s}$$

$$v = 7.5 \text{ m/s}$$

$$a = 0.60 \text{ m/s}^2$$

$$\cancel{x} =$$

$$\text{Form: } s = \frac{v^2 - u^2}{2a}$$

$$v^2 = 7.5^2 = 56.25$$

$$u^2 = 2.5^2 = 6.25$$

$$v^2 - u^2 = 56.25 - 6.25 = 50$$

$$2 \times a = 2 \times 0.60 = 1.2$$

$$s = 50 \div 1.2 = 41.7 = 42 \text{ m}$$

$$\text{Check: } v^2 - u^2 = 50 \quad 2as = 50$$

E2. A train speeds up from 15 m/s to 25 m/s over 350 m . Calculate the acceleration.

Givens

$$s = 350 \text{ m}$$

$$u = 15 \text{ m/s}$$

$$v = 25 \text{ m/s}$$

$$a = ?$$

$$\cancel{x} =$$

$$\text{Form: } a = \frac{v^2 - u^2}{2s}$$

$$v^2 = 25^2 = 625$$

$$u^2 = 15^2 = 225$$

$$v^2 - u^2 = 625 - 225 = 400$$

$$2 \times s = 2 \times 350 = 700$$

$$a = 400 \div 700 = 0.571 = 0.57 \text{ m/s}^2$$

$$\text{Check: } v^2 - u^2 = 400 \quad 2as = 400$$

E3. A van moving at 4.0 m/s accelerates at 1.2 m/s² over 50 m. Calculate the final velocity.

Givens

$$s = 50 \text{ m}$$

$$u = 4.0 \text{ m/s}$$

$$v = ?$$

$$a = 1.2 \text{ m/s}^2$$

$$\cancel{v} =$$

$$\text{Form: } v = \sqrt{u^2 + 2as}$$

$$u^2 = 4.0^2 = 16$$

$$2as = 2 \times 1.2 \times 50 = 120$$

$$v^2 = 16 + 120 = 136$$

$$v = \sqrt{136} = 11.66 = \mathbf{12 \text{ m/s}}$$

Check: $v^2 - u^2 = \underline{120}$ $2as = \underline{120}$

E4. A car accelerates at 1.8 m/s² over 60 m and reaches 20 m/s. Calculate its initial velocity.

Givens

$$s = 60 \text{ m}$$

$$u = ?$$

$$v = 20 \text{ m/s}$$

$$a = 1.8 \text{ m/s}^2$$

$$\cancel{v} =$$

$$\text{Form: } u = \sqrt{v^2 - 2as}$$

$$v^2 = 20^2 = 400$$

$$2as = 2 \times 1.8 \times 60 = 216$$

$$u^2 = 400 - 216 = 184$$

$$u = \sqrt{184} = 13.56 = \mathbf{14 \text{ m/s}}$$

Check: $v^2 - u^2 = \underline{216}$ $2as = \underline{216}$